

AWS Hosting Cost Estimate

Monthly run-rate for CliniOps on Amazon EKS with RDS for MySQL, sized for medium data volume and medium traffic. All rates are US East (N. Virginia) on-demand list prices, September 2026.

BASE MONTHLY

\$472.78

On-demand, no commitment

WITH 15% BUFFER

\$543.70

Recommended budget line

WITH 1-YR SAVINGS PLAN

\$376.49

Same architecture, committed

ANNUAL (BASE)

\$5,673

\$4,518 committed

✓ Within the \$400–500 target band, excluding buffer

01 What “medium” means here

Sizing is meaningless without a workload definition, so these are the figures every number below is derived from. If actual usage differs materially, the estimate moves with it.

Dimension	Assumed	Drives
Named users	40–60 across one organisation	App replicas, RDS connections
Peak concurrent users	20–25	vCPU headroom
Natural-language queries	~1,500 / month (~50/day)	LLM spend, CPU
Structured data in RDS	~50 GB, growing ~4 GB/month	RDS storage, instance class
Documents indexed	~15,000 files → ~400k chunks	RDS storage, ingestion CPU
Document ingestion	~10 GB/month, in bursts	Peak memory and CPU
Dashboards	~40 dashboards, ~200 widgets	Query volume on open
Internet egress	~100 GB/month	Data transfer
Availability target	Business hours, single region, 2 AZs	Node count, RDS topology

02 Why the nodes are sized the way they are

The dominant sizing constraint is not traffic — it is that CliniOps runs its embedding model **in-process**. `app/utils/embedding_utils.py` loads `intfloat/multilingual-e5-large` at import time, so every application worker holds its own copy of the weights.

This footprint was measured on the actual codebase rather than estimated:

Stage	Resident memory
Bare Python process	19 MB
+ torch / sentence-transformers imported	440 MB
+ model weights loaded	786 MB

+ one batch encode (steady state)	1,946 MB
+ faiss, pandas, numpy	1,951 MB

So roughly **2 GB per worker at rest**, with transient peaks during document ingestion when a whole file's chunks are embedded in one call. Two Gunicorn workers per pod gives a working set of about 4 GB and a ceiling around 5 GB.

Resource	Setting	Reasoning
App pod	2 replicas · request 3 GiB / 1 vCPU · limit 5 GiB / 2 vCPU	Measured footprint plus ingestion headroom
Worker nodes	2 × m6a.xlarge (4 vCPU, 16 GiB)	One app pod per node across 2 AZs
Node disk	50 GB gp3 each	Container image is large — see below
Autoscaling	2 → 4 nodes via Karpenter	Ingestion and morning load are spiky
RDS	db.t4g.medium (2 vCPU, 4 GiB), Single-AZ	Low query volume; burst credits cover peaks
RDS storage	100 GB gp3	50 GB data + chunk/embedding rows + growth

BAKE THE MODEL INTO THE IMAGE

The model cache is **2.2 GB on disk**. If it is not baked into the container image, every pod start re-downloads it from Hugging Face — slow rollouts, and NAT gateway data-processing charges on every scale-out. Build it in, and size node disk and ECR accordingly. This is why the image is budgeted at ~6 GB.

DO NOT RAISE THE WORKER COUNT CASUALLY

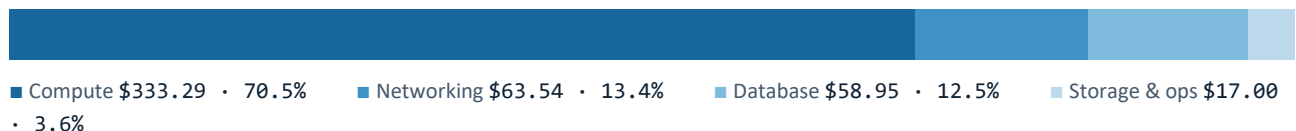
Each additional Gunicorn worker adds ~2 GB. Running `--workers 4` would put a pod at ~8 GB and force a larger instance class. Prefer `--workers 2 --threads 4 --preload`, and pin `OMP_NUM_THREADS` so torch does not oversubscribe the 4 vCPUs.

03 Monthly cost breakdown

US East (N. Virginia), on-demand, **730 hours** per month. Every line shows the unit rate and the arithmetic so it can be checked independently.

Component	Configuration	Unit rate	Qty	Monthly
Compute – Kubernetes				
EKS control plane	1 cluster, standard support	\$0.10 / hr	730 hr	\$73.00
EC2 worker nodes	2 × m6a.xlarge, Linux	\$0.1728 / hr	1,460 hr	\$252.29
EBS root volumes	2 × 50 GB gp3	\$0.08 / GB	100 GB	\$8.00
Compute subtotal				\$333.29
Database				
RDS for MySQL	db.t4g.medium, Single-AZ	\$0.065 / hr	730 hr	\$47.45
RDS storage	100 GB gp3	\$0.115 / GB	100 GB	\$11.50
Automated backups	Within free allocation	\$0.00 / GB	100 GB	\$0.00
Database subtotal				\$58.95
Networking				
NAT gateway	1 gateway, shared across AZs	\$0.045 / hr	730 hr	\$32.85
NAT data processing	Outbound from private subnets	\$0.045 / GB	100 GB	\$4.50
Application Load Balancer	1 ALB, HTTPS	\$0.0225 / hr	730 hr	\$16.43
ALB capacity units	~1.5 LCU average	\$0.008 / LCU-hr	1,095	\$8.76
Internet egress	First 100 GB/month free	\$0.09 / GB	0 GB	\$0.00
Cross-AZ traffic	Pod ↔ RDS, pod ↔ pod	\$0.02 / GB	50 GB	\$1.00
Networking subtotal				\$63.54
Storage & operations				
ECR	~6 GB image × 5 retained tags	\$0.10 / GB	30 GB	\$3.00
S3	Documents, exports, DB dumps	\$0.023 / GB	150 GB	\$3.45
CloudWatch Logs	15 GB ingest + 15 GB retained	\$0.50 / \$0.03	15 GB	\$7.95
Secrets Manager	OpenAI key, DB, SMTP, JWT	\$0.40 / secret	4	\$1.60
Route 53	1 hosted zone + queries	\$0.50 / zone	1	\$1.00
Storage & ops subtotal				\$17.00
Base monthly total				\$472.78

Where the money goes



Compute is 70% of the bill, and two thirds of that is the EC2 nodes. That is where any meaningful optimisation has to come from.

04 Cross-check against AWS pricing

Every unit rate above was checked against AWS's published pricing pages and independent trackers in September 2026. To reproduce the figure in the AWS Pricing Calculator (calculator.aws), add these services with region **US East (N. Virginia)**:

Calculator service	Inputs to enter	Expected
Amazon EKS	1 cluster, standard support, 730 hrs	\$73.00
Amazon EC2	2 × m6a.xlarge, Linux, on-demand, 100% util., 50 GB gp3 each	\$260.29
Amazon RDS for MySQL	db.t4g.medium, Single-AZ, 100 GB gp3, on-demand	\$58.95
VPC / NAT gateway	1 gateway, 100 GB processed	\$37.35
Elastic Load Balancing	1 ALB, 1.5 average LCU	\$25.19
Amazon ECR	30 GB stored	\$3.00
Amazon S3	150 GB Standard	\$3.45
CloudWatch	15 GB logs ingested and stored	\$7.95
Secrets Manager · Route 53	4 secrets · 1 hosted zone	\$2.60
Data transfer	100 GB out (free tier), 50 GB cross-AZ	\$1.00
Calculator total		\$472.78

ONE RATE TO CONFIRM AT PURCHASE

Published sources disagree slightly on `db.t4g.medium`: AWS list is **\$0.065/hr**, while some trackers report **\$0.070/hr**. At the higher rate the total becomes **\$476.43** — still inside the target band, but confirm it in the calculator before committing.

05 Buffer and variance

The base figure assumes the workload sits at the assumptions in section 01. A **15% buffer (\$70.92)** gives a budget line of **\$543.70**. The lines most likely to move:

Line	Why it might move	Plausible swing
NAT data processing	Pod restarts pulling images, OpenAI API calls, package installs	+\$10–30
CloudWatch Logs	The app logs verbosely; debug output at volume adds up fast	+\$10–40
Node autoscaling	Ingestion bursts adding a third or fourth node for hours	+\$20–60
RDS storage growth	Chunk and embedding rows grow with every document indexed	+\$5–15
Egress	Large exports or dashboard-heavy usage crossing the free tier	+\$0–20

06 Ways to spend less

Lever	Trade-off	Saving	New total
1-yr Compute Savings Plan on nodes	Commitment; rate drops to \$0.114/hr	−\$85.85	\$386.93
1-yr RDS Reserved Instance	Commitment; ~22% off compute	−\$10.44	\$462.34
Both commitments together	1-year lock-in on the stable core	−\$96.29	\$376.49
Drop to 2 × m6a.large	8 GiB nodes — no ingestion headroom	−\$126.15	\$346.63
Graviton nodes (m7g.xlarge)	Verify ARM wheels for pyodbc, faiss, torch	−\$14.02	\$458.76
VPC endpoints for S3 and ECR	\$7.30/endpoint, removes most NAT processing	−\$0-15	—
Scale nodes to zero out of hours	Only if no overnight ingestion; cold starts	−\$60-90	—

The most defensible package for a production launch is **both commitments**: it holds the architecture exactly as specified and lands at **\$376.49/month**, leaving the whole \$400–500 band as genuine headroom.

07 Not included: LLM API spend

This is not an AWS charge, but it is the largest variable cost in running CliniOps and belongs in any hosting budget. Embeddings run locally at no per-call cost; text generation goes to OpenAI.

The code currently pins the expensive model. `app/services/query_sql_agent.py` hardcodes `model='gpt-4'` for SQL generation regardless of what the agent is configured with, and agents default to `gpt-4` for synthesis. Legacy GPT-4 is roughly 20× the price of current small models.

Path	Model in code	Per query	1,500 queries
SQL generation	<code>gpt-4</code> (hardcoded)	~\$0.147	~\$221
Answer synthesis	<code>gpt-4</code> (agent default)	~\$0.060	~\$90
Follow-up suggestions	<code>gpt-4</code>	~\$0.023	~\$34
Document Q&A / routing	<code>gpt-4o-mini</code>	~\$0.003	~\$5
Estimated monthly LLM spend, as-is			~\$350

WORTH FIXING BEFORE LAUNCH

Moving SQL generation and synthesis onto a current small model brings the same workload to roughly **\$30–45/month** — a saving of around **\$300/month**, comparable to the entire EC2 line. Schema size in the SQL prompt is the other lever: the prompt currently includes every reflected table, so restricting agents to the tables they need cuts input tokens directly.

Token estimates assume ~4,600 input / 150 output for SQL generation against a 40-table schema. Confirm current list prices before budgeting.

08 If the workload changes

Scenario	Shape	AWS / month
Small — pilot	≤15 users, 1 × m6a.large, db.t4g.small, 50 GB	~\$235
Medium — this estimate	40–60 users, 2 × m6a.xlarge, db.t4g.medium, 100 GB	\$472.78
Medium + HA database	As above with Multi-AZ RDS	~\$520
Large	~150 users, 4 × m6a.xlarge, db.r6g.large Multi-AZ, 300 GB	~\$1,100

Growth is close to linear in node count until the embedding workload justifies splitting it into its own deployment. At that point the app pods shrink dramatically and only the embedding service needs the memory — worth revisiting past roughly 100 users.

09 Assumptions and exclusions

Assumptions

- Region **us-east-1**. EU regions run roughly 8–15% higher; if clinical data must stay in the EU, add that margin.
- 730 hours per month; nodes and RDS run continuously.
- On-demand list pricing, no existing Enterprise Discount Program or credits applied.
- Two availability zones, one region. No disaster-recovery region.
- Single NAT gateway shared across AZs — cheaper, but a zone-level single point of failure.
- RDS Single-AZ with automated backups and point-in-time recovery. Multi-AZ costed in section 08.
- The embedding model is baked into the container image, not downloaded at pod start.
- Uploads are processed in memory and to ephemeral temp files, so no EFS or persistent volume is required. This matches how the ingestion code works today.
- TLS certificates via ACM (no charge). Domain registration not included.
- Basic CloudWatch logging only — no Container Insights, which would add materially.

Excluded

- OpenAI API spend — estimated separately in section 07.
- Non-production environments. A scaled-down staging cluster typically adds 40–60% of the production figure; sharing one EKS cluster with separate namespaces is far cheaper than a second control plane.
- Engineering time for build, migration and ongoing operations.
- AWS Support plans. Business support is the greater of \$100/month or 10% of spend.
- Data migration from the current SQLite database, and any one-off transfer charges.
- Third-party monitoring, WAF, or security tooling.
- Sales tax and VAT.

Rates verified September 2026 against AWS EKS pricing, RDS for MySQL pricing, Elastic Load Balancing pricing and independent instance trackers. AWS prices change; re-run the AWS Pricing Calculator (calculator.aws) before contracting.